Ch. 10 — P02 (A) Sample Size and Confidence Interval Width

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A pilot study of $n_0 = 10$ replications of a SimPy simulation yields:

$$\bar{X} = 14.7 \text{ min}, \qquad s = 3.2 \text{ min}$$

Tasks

1. Compute the 95% confidence interval for the true mean. Use the t-distribution with $n_0 - 1 = 9$ degrees of freedom. Report the half-width h.
2. **Sequential planning:** The analyst wants a half-width of $h^* = 0.5$ minutes. Using the formula:

   $$n^* = \left\lceil \left(\frac{t_{9,\,0.025} \cdot s}{h^*} \right)^2 \right\rceil$$

   compute n^*.

   The analyst is surprised by how large n^* is. Explain why small h^* requires disproportionately more replications (reference the $h \propto 1/\sqrt{n}$ relationship).
3. **Run the second stage:** Suppose after running n^* replications, the analyst obtains $\bar{X} = 15.1$ min, $s = 3.0$ min. Compute the new 95% CI. Does the achieved half-width meet the target? What if the true mean is 15.5 min — is the true value covered?
4. **Cost tradeoff:** Each replication costs 2 minutes of compute time. Complete the table below and recommend the most cost-effective sample size that keeps $h \leq 1.0$ min. Use $s = 3.2$ throughout (conservative estimate from pilot).

n	h (min)	Total cost (min)
10		
20		
30		
50		
100		

5. A colleague argues: “Instead of 50 replications of 500 customers, run 1 replication of 25,000 customers — it’s the same sample size.” Identify the two key reasons this is not equivalent.